



JAMAICA CONSTABULARY FORCE

Data Centre

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1 Purpose

This Standard Operating Procedure (SOP) will serve as guiding policy document to ensure smooth operations of Jamaica Constabulary Force (JCF) Data Centre. The objective is to achieve efficiency, quality output and uniformity of performance, while reducing miscommunication and failure to comply with industry regulations. It will clarify processes by which all current and future stakeholders for Data Centre shall be mandated to follow and adhere.

2 Scope

This SOP shall apply to all Data Centre staffs, contractors, vendors, customers and all the stakeholders who are availing data centre services. This SOP shall also cover all end user devices, servers, storage, networking devices, devices being used to configure and manage service installation remotely connected to Data Centre services.

3 Introduction

Data Centre is a critical national ICT infrastructure, which serves as platform to efficiently deliver secure, reliable and real time services. It is located at Telecommunication Division. Data Centre is located within a 2500 sq.ft data center space. It is equipped with state-of-art tier2 facilities including power, cooling, rack space, fire safety and other physical security feature.

4 Role Definition

Authorized Staff/User: employees who are authorized to gain access to the Data Centre.

Authorized Vendor/: Private contractor who, through contractual arrangement and appropriate approvals, have access to the Data Centre.

Data Center Staff: JCF employees who work at the Data Centre

Visitors: Any other personnel who may occasionally visit Data Centre.

5 Physical Access

5.1 Staff/Vendor

Authorizations will only be approved for individual(s) who are responsible for installation and/or maintenance of equipment housed in the JCF Data Center. Approval processes are as follows:

1. Concerned staff/vendor must request access and submit to Telecoms Operations Officer for authorization.
2. Authorized vendors shall be allowed entrance into the Data Centre area by a Data Center employee. Authorized vendors will operate in the presence of a Data Centre Staff.
3. Authorized staff/vendors are responsible for logging in/out when entering/exiting the Data Centre. The purpose of the visit must be documented. See attached prescribe form (Annexure A).

5.2 Visitor

Anyone who is not a Data Centre employee, an authorized staff member, or authorized vendor is considered a visitor. All visitors to the Data Center must adhere to the following procedures:

1. Visitors must be accompanied by either a Data Center employee or other authorized staff member at all times while in the Data Center.
2. Visitors must log in/out when entering/exiting the Data Center. The purpose of the visit must be documented.
3. Visits should be scheduled through the appropriate Officer. Unscheduled visits to install equipment or perform other tasks *may* not be entertained without prior notification.

6 Remote Access

6.1 Virtual Private Network (VPN) and Secure Shell Access (SSH)

Only approved employees and authorized third parties (customers, vendors, etc.) shall be provided VPN clients credentials. Only one VPN access credentials will be provided.

Authorized employees or third parties must adhere to following requirement:

1. Fill up form or make request via email and submit to Officer in Charge. See Appendix C. This is recommended to be done at least 24 hours prior to the time access is needed.

2. It is the responsibility of employees inclusive the security team to ensure that only unauthorized users are allowed access to the network.
3. When actively connected to the corporate network, VPNs will force all traffic to and from the PC over the VPN tunnel.
4. Dual (split) tunneling is NOT permitted; only one network connection is allowed.
5. VPN users will be automatically disconnected from network after thirty minutes of inactivity. The user must then logon again to reconnect to the network.
6. The VPN concentrator is limited to an absolute connection time which is dependent on the scope of work to be done.
7. Only Data Centre approved VPN clients must be used.
8. Authorized user shall protect their username and passwords even from family members.
9. While using a computer to remotely connect to Data Centre network, Authorized Users shall ensure the remote host is not connected to any other network at the same time, with the exception of personal networks that are under their complete control or under the complete control of an Authorized User or Third Party.
10. All hosts that are connected to internal networks via remote access technologies must use the most up-to-date anti-virus software.

7 Environment and Safety

The Data Centre is equipped with dual air condition units which presently work simultaneously at an agreed temperature. This is subject to change based on the number of equipment and the heat generating from those equipment as additional devices are added/removed. The data center now has two rows of cabinets should a third row come into effect a third air conditioning unit will be required.

A General Electric 30 kva uninterrupted power supply is installed in which a second one is to be procured to provide redundancy.

Fire suppression system was installed based on the size of the room. There are 6 smoke detectors, 2 pull manual stations, 2 abort switches, 1 pressure switch, 2 horn and strobe with audible, 1 horn and strobe without and a UPS panel. Any two smoke detectors triggered will cause the fire suppression to become activated. Any loud noise or work involving dust that may take place within the Data Centre, upon approval Operations Officer, should have the fire suppression deactivated and re-activate immediately upon completion of that particular work.

Cable trays separating power from data. The separation of power and data is to be maintained.

Environmental Monitoring: Geist Watchdog software monitors software temperature, relative humidity, light level, air flow, sound, leak rope. An agreed reading is set and any deviation may trigger an alert; in some cases an increase or reduced reading. The agreed amount can only be changed by the Operations Officer or a senior manager.

Be careful not to step on the leak rope it may become damage.

Close circuit camera monitors the Telecommunication Division and its environ inclusive of the data centre. Entrance into the Data Centre required electronic door access (fingerprint).

There are two doors. The main door is used for enter and exit of the data centre. It is operated by a biometrics device. User enter with either pointer fingers once that users has been configured into the system by an authorized administrator. The door will automatically open. Persons without biometrics access is required to log their attendance on the prescribed sheet prior to entry. The second door is for exit purposes and is only use as an entrance door upon receipt of large hardware delivered for the data centre. The second door is accessibly only from inside.

What to do if where a Fire, Earthquake or Flood occurs.

- Fire – See Appendix E
- Earthquake –
 - The Structure is so that the building will remain standing for certain categories of earthquake
 - Network Cabinets/ Racks are fasten to the ground with most devices adequately racked and bolted to prevent fallen to the ground
 - Stand in the door jam until the shaking stops

- After the shaking ends exit the building and proceed to the assemble area for administrative routine
- Flood – The data centre is above ground level hence flooding will take some time before water enters the data centre. The leak rope will alarm alerting of water in the room. Where possible power down systems and equipment and cause equipment to be moved to higher floor and inform the immediate the immediate supervisor as well as relevant superiors.

After an emergency event all services and equipment must be checked to ensure they are working properly

8 Resources

8.1 Hardware/Equipment

Hardware/Equipment refers to all devices (storage, computer, server, network), racks, cables installed within the Data Centre designated area. All installation, replacement and removal works shall be approved by Operations Officer.

Authorized staff/vendor performing equipment installation must submit a Change form(Appendix B) and get approval before carrying out any of the following tasks.

1. Installation of new hardware.
2. Replacement of hardware.
3. Removal/Decommissioning of defective equipment or expired equipment.

8.2 Application/Software

Application/Software refers to all proprietary, in-house or outsourced software as well as all open source software being used to run the services at Data Centre. All installation, configuration, testing, upgrades, updates and removal of application/software must be approved by Operations Officer in consultation with concerned agency.

The authorized staff/vendor performing installation or configuration works shall submit a Change form (Appendix B) and get approval before carrying out any of the following tasks:

- Installation and configuration.
- Upgrades/Updates.
- Decommissioning/Uninstallation/Removal.

8.3 Allocation

Resource refers to all usable network, compute and storage capacities currently available and ready to be allocated. The concerned agency wishing to avail resources at Data Centre must follow procedures outline below:

Seek approval from the Operations Officer to host services/application at Data Centre using the Change Form. The application must be accompanied with a comprehensive project document in case of new application/system/website.

Seek to conduct thorough testing prior to implementing into the production environment.

9 Reporting and Documentation

A report must be compiled and submitted to the Operations Officer regarding routine operations or happenings concerning the data centre on a weekly and monthly basis or as often as required/necessary.

All network communication logs must be maintained for a period of six months and archived for a period of two years.

The monitoring of daily activities should be recorded and stored for a period of 2 years.

All installation/change/decommission of any equipment or software must be captured in the inventory and appropriate documented and kept for a period of no less than three years.

10 Human Resources

10.1 Training

Capacity building is key to ensure smooth operation of Data Centre. Technical team shall be provided trainings in following field.

No.	Category	Modules	Target Audience
1	Advance System Administration	1 Virtualization 2 Storage 3 Operating Systems (DNS, DHCP, Log Server, Network Management Software, Network Time Protocol)	Data Centre Staff
2	Data Centre Network Security	Firewall Router Switch	Data Centre Staff Security Staff Network Staff
3	Electrical	Basic Electrical	Data Centre Staff, Network Staff
4	Application Database	Introduction to Databases	Data Centre Staff Database Administrator

There needs to be sufficient capable staff to monitor the data centre in the event where a Staff member goes on leave.

10.2 Roles and Communication

Personnel	Composition	Responsibilities
Management	ACP Technology Branch Commanding Officer Telecom	Overall responsibility of the divisions to include the data centre Provide direction and guidance in line with the strategic direction of the JCF. Authorize and Approve Projects Scope with regards to change management, new implementation and decommissioning Forecast and manage data centre resources
Project Management Unit		Prepare and plan project scope based on available resource

		Plan and coordinate
Application Owner	Comanding Officer of the Division requesting the application	Owner of the data housed in the in the Data Centre
Operations	Data Cente Technicians	Manage and apply patches to servers Create virtual space to host applications (processor, memory, storage, network) Plan and outline patch management Configure manage and maintain servers Provide 24/7 or 9/5 on-call support, as specified for each supported server or device; diagnose and correct hardware and operating- system problems; provide support to application and database administrators when they are troubleshooting applications/services Provide unique usernames and passwords for all virtual platform and applications/services to Operations Officer

		Configure, maintain and monitor servers such as DNS, Log Server, NTP, NMS and other critical servers required to run and operate Data Centre network and services.
System Administrator	System Administrator	Monitor and maintain the application running within the data centre
Security	It Security Staff	Provide support in ensuring the physical and electronic data is kept safe
Network Department	Network Staff	Provide support in ensuring only authorized access is granted and unauthorized access is kept out

12 Inventory Management

An inventory of all hardware/software/application shall be maintained.

12.1 Hardware

An up-to-date list of hardware component in Data Centre shall be maintained to include warranty and maintenance information.

12.2 Software Library

An up-to-date list of applications hosted in Data Centre shall be maintained to include warranty and maintenance information.

12.3 Service/Applications Catalogue

A list of software licenses/versions, Operating Systems and other critical software shall be maintained to include warranty and maintenance information.

13 Configuration and Testing

13.1 Operations Officer shall provide a staging server where new software/applications shall be temporarily hosted for testing before going into production environment.

13.2 All configuration and testing of hardware/software shall be carried out in the staging server.

13.3 Staging server will be a temporary provision until the end of testing phase. After completion of tests, all resource shall be held back and all configurations shall be erased.

12.4 A test documentation must be completed to include reason, test conducted, result, risk, lessons learnt etc.

13 Monitoring Activities of the Data Centre

13.1 The Data Centre is to be monitored 24 hours daily

13.1 Daily monitoring will include but not confined to physical environment, storage, servers.

13.2 Data Centre Staff is required to physically check the room twice daily 8:45 am and 3:00 pm.

13.3 Data Centre Staff is required to monitor most activities from the designated Monitoring Room and capture monitored information on a log sheet (Appendix D).

14 Cleaning of the Data Centre

14.1 The Data Centre shall be cleaned twice weekly.

14.2 The cleaning agent shall not comprise of strongly scantly chemicals

14.3 No water is to be thrown/poured on the floor during cleanup nor a bucket with cleaning liquid enter the datacentre.

14.4 Personnel cleaning must be aware of the nature of the operations and as such the importance of having a clean and room within the stipulated requirement.

14.5 A scheduled maintenance incorporates servicing, dusting is to be conducted

15 Glossary

Data Centre Data Center

NMS Network Monitoring System

VPN Virtual Private Network

15 Annexure

Data Centre Entry and Exit Log

Data Centre Daily Check Sheet

Service Log Form

Change Request Form

User Access Form



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Data Centre Entry and Exit Log							
Date	First Name	Last Name	Company	Purpose	Signature	Time In	Time Out



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Data Centre Daily Check Sheet

Vsphere

	Capacity	Available		Count
CPU			Host	
Memory			Virtual Machine	
Storage			Cluster	
			Network	
			Datastore	

NAS

Drive Bay 1	1	2	3	4	5	6	7	8	9	10	11	12
Health Check												
Drive Bay 2												
Health Check												
				Capacity				Available				
Hxbackup				103.7 TB								
Archive				102.8 TB								

<u>Nodes</u>	<u>Online (√)/Offline(X)</u>	<u>UPS</u>	<u>Load Capacity</u>
Node 1		L1	
Node 2		L2	
Node 3		L2	
		Battery Charge Level %	

Checked By:	Signature:	Date and Time:
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<u>Physical Environment</u>	Normal(√)/Abnormal(X)	Remarks
Temperature		
Humidity		
Lighting		

Standard Operating Procedures for the JCF Data Centre

JCF/TECB/PR/SSs/0002/2022

Email: technologybranch@jcf.gov.jm or 948tech@jcf.gov.jm

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Appendix E: Fire Response

Autopulse 542R Conventional Suppression Control & Fire Alarm System Operating Instructions

Normal Standby Operation

1. Green "Power On" LED lit steadily
2. Red "Alarm" LED off
3. Yellow Trouble & Supervisory LED's off

Alarm Condition

Upon receipt of an alarm from one activated detector in one zone:

1. Red Zone 1 or Zone 2 Alarm LED Lit
2. Sounder on
3. Signal circuit device on for 1 second every 2 seconds

Upon activation of an alarm from a second activated detector in the other zone:

1. Red "Alarm" LED Lit
2. Sounder on
3. Signal circuit devices on steady
4. Programmed release time delay activated

Upon receipt of an alarm from an activated manual station:

1. Red Manual Station Alarm LED lit
2. Sounder on
3. Signal circuit devices on steady
4. Releasing circuit will ACTIVATE

Operation of ABORT:

1. Yellow ABORT ACTIVATED LED Lit
2. Discharge is aborted ONLY when alarm is from a detector and abort is activated during programmed release time delay. Abort sequence is dependent upon programming

Alarm Acknowledge & Silence

To Acknowledge an Alarm state and silence the Audible alarm, press the "ALARM SILENCE" Switch. The Alarm Silence LED will be on steady.

System Reset

Press the system "RESET" switch and the system will reset if all the fault conditions are now clear.

Trouble Conditions

Activation of trouble signals under normal operation indicates a condition that requires **immediate** attention. Contract your local service representative. The system audible indicator will signal the trouble tone and trouble LED Indicator will flash. The system audible indicator may be silenced by pressing the "TONE SILENCE" switch. When pressed the audible will silence and the trouble LED will go on steady.

Switch Functions

- 1) **TONE SILENCE** – This switch when activated acknowledges that there is active alarm, supervisory or trouble conditions on this panel.
- 2) **ALARM SILENCE** – This switch when activated will silence all active signal output circuits.
- 3) **ALARM ACTIVATE** – This switch maybe used to activate Notification Appliance Circuits.
- 4) **RESET** – When all active alarm events from this panel have been cleared depressing this switch:
 - Turns off Alarm LED
 - Returns system to normal standby state, if no active supervisory or troubles conditions exist on the system.

Maintenance & Testing

To ensure reliable and proper system operation inspection of the system should be scheduled in accordance with NFPA 72 -1993 and Local Fire Codes. A qualified factory trained service person should perform testing

Prior to testing: Notify fire department and/or central alarm receiving station if alarm signal is transmitted. Notify all facility personnel of the test so alarm sounding devices are ignored during the test period.

After Testing: Notify all fire, central station and building personnel when test has been completed.

LOCAL SERVICE REPRESENTATIVE



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